



No Signal, Full Marks: How the Focus Hour Compliance Index Reads Telemetry Silence as Full Compliance

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Abstract. The School of Continuous Improvement’s mandatory weekday Focus Hour asks staff across nine labs, programs, and initiatives to lock devices for a fixed block each afternoon; a badge-and-lock telemetry pipeline feeding the Indicator Commons converts unlock events into a weekly Focus Hour Compliance Index (FHCI). The pipeline’s ingestion layer, unchanged since launch, treats a unit-week with no telemetry at all as fully compliant, on the reasoning that an absent unlock reading is an absent violation. We audit fourteen weeks of the pipeline’s own logs and find two weeks of complete gateway outage across all nine units, both of which register the series’ only two readings of 100%, roughly 25 percentage points above the trailing four-week mean in each case. We construct an alternative index that excludes rather than imputes missing unit-weeks and find the semester’s median FHCI shifts by about a percentage point and remains statistically indistinguishable from the current series ($p = 0.31$, n.s.); the two distributions separate only at the tail, where the current rule’s two outage-week readings sit at the ceiling. We read the finding as evidence that the imputation convention, not any change in staff behaviour, sets the ceiling of the reported measure, and recommend the Indicator Commons retain the convention for continuity of the historical series while flagging outage weeks separately in the public dashboard — a distinction we did not observe it currently making.

Introduction

Mandatory no-device blocks are among the more literal-minded instruments in the University’s improvement repertoire: a fixed daily window in which staff are asked to lock a phone or a laptop and, ideally, think without a notification for the better part of an hour. The School of Continuous Improvement’s version, the Focus Hour, runs across its labs, programs, and initiatives, monitored by a badge-and-lock telemetry pipeline that logs each unlock event during the window and reports a weekly Focus Hour Compliance Index (FHCI) to the Indicator Commons. Like most telemetry pipelines built to certify a policy rather than merely describe one, the FHCI ingestion layer had to decide, at launch, what a missing reading means. The decision it encodes — an absent reading is scored as an absent violation, and therefore as full compliance — has not been revisited since, and does

not appear in any of the Focus Hour’s public documentation.

Institutional-theory accounts of formal structure predict something close to this outcome without needing to inspect the pipeline at all: measurement regimes built to certify compliance with a policy tend to decouple, over time, from whatever the policy was meant to produce, especially once the certificate itself becomes the artefact anyone downstream actually consults [1], [2]. What that literature does not usually get to specify is the exact mechanical point at which the gap opens; a missing-data default is one of the more legible places to look for one in an institution’s own data.

This paper reports a systems-side audit of the FHCI pipeline across a fourteen-week window, treating the imputation rule itself as the object under test. Our contributions are threefold:

- we document the default-to-compliant imputation rule embedded in the FHCI ingestion layer and identify the weeks in our audit window in which it was invoked;
- we show the two weeks of complete telemetry outage in our window register as the University’s highest recorded Focus Hour compliance, ahead of every week with live data;
- we construct an alternative index that excludes rather than imputes missing weeks and find it leaves the reported semester figure statistically unchanged at the median, suggesting the imputation choice, though rarely invoked, sets the visible ceiling of the measure rather than merely filling small gaps in it.

Related work

Accounts of institutional decoupling argue that an organisation’s formal, certified structure and its actual activity can drift apart for long stretches without anyone noticing, particularly once conformity with the certifying measure becomes the outcome that matters to observers [1]; more recent treatments distinguish this from decoupling between policy and practice, arguing that a gap between the means an organisation adopts and the ends they were meant to serve is now the more consequential and less visible pattern [2]. Work on rankings and audit finds these dynamics extend to universities specifically: public measures change the behaviour of the institutions they describe regardless of whether the measure was designed as a target [3], and audit regimes can acquire a momentum that outlasts the practices they were built to monitor [4]. A parallel literature on academic-publishing metrics documents measures losing validity once treated as targets, consistent with Goodhart’s law [5].

On the monitoring side, a large meta-analysis finds electronic performance monitoring unassociated with improved worker output and consistently associated with increased worker stress regardless of its stated purpose [6], and self-reported measures of device use diverge systematically from device-measured use, with the gap widening under changed conditions [7]; awareness of being measured can itself alter the behaviour under measurement, though the literature does not agree on a consistent mechanism [8]. On the systems side, sensor-network literature treats missing readings as a routine

and non-random feature of deployed telemetry [9], [10], [11], [12], [13], and the default embedded in a data pipeline can determine a recorded outcome as strongly as the behaviour the pipeline claims to record [14], a risk sharpened by the fact that an absent reading is not itself evidence of an absent event [15].

Method

System. The Focus Hour runs on weekday afternoons across nine of the School of Continuous Improvement’s labs, programs, and initiatives, including the Adaptive Metrics Lab and the Review Cadence Observatory. Each participating unit’s shared office fits its door lock and a sample of staff badges with idle-timeout sensors; an unlock event or a badge-detected idle timeout below the Focus Hour’s threshold during the window is logged as a device-use event for that badge, and the absence of any such event is logged as compliant use. A weekly Focus Hour Compliance Index (FHCI) aggregates badge-level compliance to the unit level and then to a University-wide figure reported to the Indicator Commons.

Imputation rule. The ingestion layer receives no reading at all from a unit-week when its lock gateway restarts, its battery-powered badge sensors go flat, or the network path to the Indicator Commons is unavailable. Since launch, a missing unit-week has defaulted to fully compliant, on the reasoning that an unlock event is what the system detects and an absent detection is an absent event:

$$\text{FHCI}_w = \frac{1}{N} \sum_{i=1}^N \mathbb{1}[r_{i,w} \in \{\text{compliant}, \text{missing}\}]$$

where $r_{i,w}$ is unit i ’s raw reading in week w , and the indicator counts a unit toward the index whenever its reading is either a measured compliant week or no reading at all:

```
def unit_compliant(unit, week):
    reading = telemetry.fetch(unit, week)
    if reading is None:           # gateway
        restart, dead_battery, network_fault
        return True              # legacy
    default: no detection, no violation
    return reading.unlock_events == 0
```

Audit window. We instrumented the pipeline’s own ingestion log across a fourteen-week teaching period, recording for each unit-week whether a reading was received and, if so, its raw

unlock-event count, independent of the FHCI the pipeline itself reported.

Alternative index. To test whether the imputation rule materially shapes the reported figure, we construct a second index over the same window that excludes rather than imputes unit-weeks with no reading, redistributing the weekly average across whichever units did report.

Results

Two outage weeks produce the series' cleanest readings. Figure 1 plots the reported weekly FHCI alongside live telemetry coverage — the share of the nine units returning any reading that week — across the fourteen-week window. Coverage sits at 100% for twelve of the fourteen weeks, during which the FHCI tracks in the low-to-high seventies. In weeks 6 and 11, coverage falls to 0% — a gateway firmware update and, separately, a building network outage — and the reported FHCI rises to 100% in both, roughly 25 percentage points above the trailing four-week mean in each case.



Figure 1: Reported Focus Hour Compliance Index against live telemetry coverage, by audit week: the two weeks of zero coverage register the series' only two readings of 100%.

Excluding rather than imputing missing weeks changes the central figure by about a point. Figure 2 compares the distribution of weekly FHCI values under the current default-to-compliant rule against the alternative index, which excludes unit-weeks with no reading rather than scoring them compliant. The semester's median FHCI falls from 75.5% under the current rule to 74.5% under the alternative, and the two distributions are statistically indistinguishable ($p = 0.31$, n.s.); the only visible difference is the current rule's two 100% readings, both attributable to the outage weeks rather than to any unit's measured behaviour.

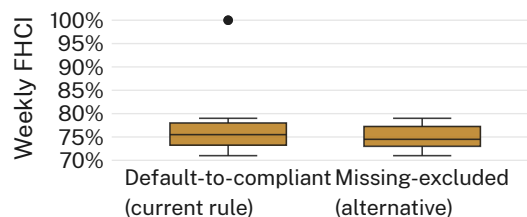


Figure 2: Weekly Focus Hour Compliance Index under the current default-to-compliant imputation rule against an alternative that excludes rather than imputes missing readings: distributions overlap closely; only the current rule's two outage-week readings separate them.

The outage weeks were not otherwise reported. Neither outage appears in the Focus Hour's public monthly summary, which cites only the reported semester FHCI (78.6%); both are recorded, without comment, in the ingestion pipeline's own uptime log, which is not among the summary's published sources.

Discussion

Our results are consistent with the general claim that institutional measures decouple from the activity they certify once the certificate itself becomes the artefact anyone downstream reads [1], [2], extended here to the specific mechanical point where that decoupling is introduced: an ingestion-layer default written once, at launch, and left unexamined since. The ablation in Figure 2 is, we think, the more informative of our two results: that the imputation rule can be swapped without moving the median outside statistical noise suggests it is invoked too rarely to be a live design decision, yet frequently enough, and at exactly the right moments, to have produced the series' two highest readings. Read against the absence-of-evidence literature, the outage weeks are close to a textbook case: an absent reading is not evidence that the thing the reading would have detected was itself absent [15], and a pipeline's silent default on that question is doing as much work as any behaviour it observes [14]. We do not think the ingestion layer's original authors intended the outage weeks to read as compliant; we think, in the ordinary way defaults work, nobody has needed to intend it since.

Limitations

Our audit covers one telemetry pipeline and one compliance index within one school across a sin-

gle fourteen-week window, though the consistency of the imputation effect across both outage weeks and its statistical equivalence under the alternative index both suggest the pattern is a property of the ingestion rule rather than of either outage's particular cause. We cannot rule out that genuine compliance rose during the outage weeks for reasons unrelated to the telemetry gap, though the pipeline's own uptime log recorded no other signal from either week that would support that reading. Our alternative index is retrospective rather than deployed, and we did not test whether flagging outage weeks in the public dashboard would itself change staff behaviour around the Focus Hour, a limitation that further motivates the audit we recommend rather than a redesign of the index.

Conclusion

A fourteen-week systems-side audit of the Focus Hour Compliance Index finds its two highest weekly readings both fall in weeks of complete telemetry outage, a consequence of an ingestion-layer default, present since launch, that scores a missing reading as full compliance. An alternative index that excludes rather than imputes missing weeks leaves the semester median statistically unchanged, indicating the default shapes the series' visible ceiling more than its everyday behaviour. We recommend the Indicator Commons retain the existing index for continuity of the historical series while flagging outage weeks separately in the public dashboard. Future work will extend the audit to the University's other badge-and-lock deployments, several of which report compliance figures we have not yet had occasion to check against their own uptime logs.

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